

Lockheed Martin Supersonic Airliner – Technical/Engineering Report

Supersonic commercial travel is re-emerging as a strategic goal. Technical studies (e.g. NASA's SCR and NRC reports) emphasize that supersonic transports require breakthroughs in propulsion, structures, avionics and systems to meet environmental and performance targets [4†L84-L93] [25†L759-L768] . Lockheed Martin's expertise (Skunk Works experience with SR-71 and X-59) and modern digital design tools can be leveraged to develop a feasible supersonic airliner. This report surveys current technology and requirements and outlines a phased development roadmap with key risks and costs.

1. Literature Survey on Supersonic Transport Design

- **Historical Context:** Concorde (Mach ~2.0) and Tu-144 were the only civil SSTs to fly passengers. Concorde achieved Mach 2 cruise but suffered high fuel burn, loud noise (afterburners at takeoff), and high maintenance costs that made operations uneconomic [34†L36-L40] [35†L43-L51] . Early studies (e.g. RAE, NASA SCR programs) identified aerodynamic “area rule” shaping, slender fuselages and highly swept, thin wings as necessary to minimize supersonic drag and sonic boom [17†L68-L77] [35†L43-L51] . Modern analyses use high-fidelity CFD and optimisation; Boom's XB-1 demonstrator used thousands of design iterations to balance low-speed stability with supersonic efficiency [34†L80-L89] .
- **Propulsion/Emission Findings:** Supersonic engines operate at higher temperatures and altitudes, so emissions (especially NO_x) and high TSFC are critical challenges [4†L84-L93] [4†L110-L118] . NASA/NRC find that while existing turbofan technology can deliver Mach 1.6 (supersonic business jet), achieving efficient Mach 2+ cruise would require *variable-cycle engines* or other advanced turbines to meet fuel-consumption and noise goals [4†L100-L107] . Boom's concept (cruise Mach 1.7) relaxes some requirements: their design uses four high-thrust, medium-bypass turbofans (no afterburners) to cruise efficiently while meeting noise standards [37†L161-L164] [35†L43-L51] . Across the literature, sustainable aviation fuel (SAF) is seen as essential: Overture is designed for 100% SAF enabling net-zero carbon operation [16†L31-L39] [27†L66-L70] .
- **Materials/Structures:** Supersonic flight (especially above Mach 1.5) generates significant skin heating and thermal stress. Modern composites (carbon-fiber laminates) and titanium alloys are preferred for their strength-to-weight and thermal stability [25†L764-L772] [31†L42-L50] . Boom's demonstrator (XB-1) uses a carbon composite fuselage with titanium bulkheads; Lockheed's SR-71 famously used titanium for its skin. Advanced studies highlight use of composites to shape aerodynamically optimum “coke-bottle” fuselages and thin wings, while managing thermal expansion [25†L764-L772] [31†L42-L50] .
- **Systems Integration:** Contemporary SST design is inherently multi-disciplinary. Leading reports stress *model-based systems engineering* (MBSE) and integrated digital platforms. Boom, for example, adopted Dassault's 3DEXPERIENCE to share parametric models across subsystems and suppliers [16†L54-L62] [16†L80-L88] . Lockheed would similarly leverage MBSE and common

data environments to synchronize aerodynamics, propulsion, control systems and safety analysis.

2. Global Regulatory and Environmental Requirements

- **Noise and Sonic Boom Regulations:** Supersonic overland flight is currently banned under ICAO rules and 14 CFR 91.817 due to loud sonic booms. However, quiet-supersonic research (NASA's X-59 QueSST) and government action are enabling new standards [6†L355-L363] [8†L333-L342]. ICAO's CAEP has agreed on recommended noise standards for future SST takeoff/landing, reflecting supersonic aircraft's unique profile [35†L13-L21]. In the US, an FAA NPRM (July 2026) proposes rescinding the Mach 1 ban and introducing a sonic-boom standard (overpressure ≤ 0.11 psf) for certified aircraft [9†L311-L319]. Boom's data (XB-1 tests) helped this effort; U.S. regulators now plan noise-based certification standards for supersonic flight [9†L311-L319] [35†L55-L63]. Lockheed must plan to meet these evolving criteria: engines and airframe must pass rigorous LTO noise tests (current Stage V equivalent) and demonstrate low boom at cruise.
- **Environmental Emissions and Climate:** The aviation sector targets net-zero CO₂ by 2050 (ICAO "LTAG"). Supersonic craft fly higher (~50–60k ft), affecting stratospheric ozone and contrails. NASA notes that more efficient propulsion (lower fuel burn) is vital because supersonics carry ~50% more fuel per passenger than subsonic for similar routes [4†L110-L118]. Use of 100% SAF is proposed as the primary mitigation: Boom aims Overture to use only SAF from first flight [16†L31-L39] [27†L66-L70]. Additionally, flying above 60k ft may reduce contrail formation (per Boom) [27†L97-L105], but increased NO_x and water vapor effects need study. Lockheed's development should include life-cycle emissions modeling, contrail avoidance flight planning, and R&D on non-CO₂ impacts as part of environmental compliance [27†L97-L105] [4†L110-L118].
- **Certification Standards:** A future supersonic airliner must meet or adapt FAR Part 25 (transport-category) requirements, plus any supersonic-specific addenda. Likely certification issues include: high-altitude aerodynamics, acoustic loads, fuel system (for SAF), and emergency procedures for high-altitude operation. Close coordination with FAA/ICAO (using X-plane data) will be needed. Boom/OHT LCE have demonstrated that data-driven testing (e.g. community noise surveys, engineering simulation) can inform regulators. Lockheed should engage regulators early (per NASA Quesst mission) to shape standards and pre-certify technologies like quiet airframes and SAF use [15†L323-L332] [35†L43-L51].

3. Propulsion System Analysis

- **Engine Architecture:** Concorde's four Olympus turbojets used afterburners during acceleration; Overture will use a modern medium-bypass turbofan ("Symphony") without afterburners [35†L43-L51]. This design emphasizes fuel efficiency and lower noise: by eliminating reheat at takeoff and cruise, takeoff noise is comparable to today's subsonic jets [35†L43-L51]. The turbofan's bypass ratio (estimated ~1:1) is a compromise allowing sufficient thrust at supersonic speed without excessive engine size. Lockheed's task is to ensure this engine achieves the required thrust (~40,000 lbf per engine) while meeting TSFC and emissions targets [37†L161-L164].
- **Variable-Cycle/Afterburner Use:** Although current Boom plans avoid afterburners, literature notes that some acceleration through Mach 1 may require spikes in thrust. Concorde fired

reheat for ~10 min to pass transonic drag [23†L293-L300] . Boom's variable-noise thrust management system may modulate throttle to minimize noise; Lockheed may consider a similar "afterburner on demand" if needed. However, NASA analyses indicate that advanced fixed-cycle engines (with efficient nozzles) could achieve Mach 1.7 "supercruise" without reheat [23†L208-L216] [25†L759-L768] , which is the Boom approach.

- **Emissions Compliance:** The engines must meet (or exceed) ICAO CAEP emissions standards (CO, HC, NO_x, smoke) at supersonic cruise altitudes. As NASA notes, higher combustion temperatures increase NO_x yield [4†L110-L118] . Engine design should incorporate lean-burn combustors or staged combustion to reduce NO_x. Use of 100% SAF inherently cuts lifecycle CO₂ by ~80–100%; Lockheed should ensure the engine is certified for full SAF operation (exploring catalytic after-treatment is likely impractical in flight). Rolls-Royce and Boom emphasize partnerships on propulsion R&D for efficiency and net-zero goals [25†L759-L768] [25†L772-L780] .
- **Intake and Exhaust:** Supersonic inlets and nozzles are critical. Boom's XB-1 uses Mach-inlet ramps to slow airflow to subsonic speeds for the engine [34†L94-L96] . Overture will require full-variable inlet geometry to efficiently decelerate air from M1.7 to ~M0.5 before compression, and a convergent/divergent nozzle to expand exhaust at cruise. Lockheed's design must minimize inlet and exhaust drag while ensuring stability across the flight envelope. Adaptive control of inlet ramps and thrust is expected (e.g. the "Variable Noise Reduction System" for takeoff) [35†L43-L51] .

4. Aerodynamics and Airframe Design

- **Wing and Fuselage:** Supersonic lift and drag requirements drive a highly-swept, thin wing (Boom's delta wing or modified compound-delta) and very slender fuselage. To minimize wave drag, designs use the Whitcomb area rule and "Coke bottle" (anti-shock) shaping: Boom's fuselage tapers from a wide cross-section near wing to a slimmer tail [17†L86-L95] [17†L111-L119] . Lockheed should employ area-ruled fuselage and planform geometry (low thickness-to-chord ratio, ~4–5%) to reduce supersonic drag [17†L86-L95] . Boom's result is a shape that is efficient at M1.7 yet stable at subsonic speeds.
- **Low-Speed Performance:** As [Boom notes](#), supersonic wings are poor for takeoff/landing. High-speed configurations need large, effective flaps and leading-edge devices. Control surfaces must be sized to ensure safe stall margins at Mach 0.3–0.4 approach speeds. Boom's design uses multiple flap segments and vortex generators for added lift in takeoff. Lockheed will need a multi-mode wing (analogous to Concorde's double-slotted flaps) and robust flight-control laws for both regimes. Lockheed can leverage F-22/F-35 control expertise for high- α flight.
- **Aerodynamic Testing:** Extensive CFD and wind-tunnel testing are planned. Boom's XB-1 tests validated computational predictions; Lockheed should similarly validate models with scaled prototypes and high-Reynolds wind-tunnel data. The NASA Quesst program also illustrates that shaped sonic boom signatures depend sensitively on nose shape and volume distribution [15†L323-L332] . Lockheed should use X-59/Quesst data to refine low-boom shaping.
- **Cruise Altitude and Performance:** Mach 1.7 cruise ~60,000 ft requires pressure cabin (~0.2 atm). Aerodynamic heating at that altitude is modest (skin ΔT ~50–100 °C above ambient), but cumulative. Materials and insulation must accommodate this (see Materials section). Drag-rise in transonic (M0.9–1.0) is mitigated by swept wings. Lockheed's envelope will likely exceed Mach 1.4–1.5 in level flight; stability margins and ascent profiles must be engineered to pass

through the transonic “sound barrier” efficiently (Boom’s data show an ability to go from M0.94 to 1.7 within constraints [35†L63-L69]).

5. Materials and Structures

- **Primary Materials:** Carbon-fiber composites (CFRP) will form the bulk of the airframe: fuselage shell, wing skins, and control surfaces. CFRP offers high stiffness-to-weight and lower coefficient of thermal expansion (CTE) than aluminum [31†L42-L50] . Boom has proven CFRP use in XB-1; modern autoclave or out-of-autoclave curing techniques can be scaled for Overture. Titanium alloys are needed in high-temperature or high-load areas (e.g. engine pylons, aft fuselage sections, landing gear bulkheads) [31†L69-L78] . Titanium’s thermal compatibility with CFRP is an advantage (similar expansion rates) [31†L69-L78] . Aluminum-lithium alloys may be used sparingly for subsonic fairings or interior fittings (for cost savings), but primary structure should avoid standard aluminum (to save weight) [25†L764-L772] .
- **Thermal Considerations:** Although Mach 1.7 generates less heating than hypersonic speeds, skin friction and compression heat raise surface temperatures (on the order of 50–100 °C above ambient) at cruise altitude. Material layups must account for this: composite resins should remain within glass-transition limits (~80–120 °C) throughout cruise. Lockheed should consider hybrid designs (e.g. thin thermal barriers or reflective coatings on leading edges) to protect resin. NASA studies of SST materials emphasize thermal stability; advanced materials like composite matrix ceramics (as mentioned for X-59 nose) could be incorporated in extreme cases [31†L33-L42] .
- **Structural Strength:** The airframe must endure both aerodynamic loads and high-altitude pressure cycles. Like subsonic transports, Overture will likely use fail-safe/fail-passive structural design. Lockheed’s F-35 program expertise with composites and damage tolerance will be valuable. Explicit tests (fatigue, impact, lightning) must be done on composite joints and carbon skins. Supersonic certification will demand proof of extended lifespan despite thermal cycling.
- **Manufacturing Techniques:** The use of digital design facilitates automated fabrication. Large composite mandrels and automated fiber placement (AFP) can reduce man-hours. Boom is building a “Superfactory” in NC with digital tooling; LM can plan similar state-of-art facilities. Additive manufacturing (3D printing) will also be used: Boom’s XB-1 used 3D-printed Ultem for brackets [31†L99-L107] . Lockheed should exploit metal-additive (e.g. DMLS titanium) for complex fittings and FAA-approved thermoplastics for non-critical interiors, to speed iterations and reduce weight.

6. Systems Integration and Supply Chain

- **MBSE and Digital Continuity:** A fully integrated digital development environment is critical. All subsystems (avionics, propulsion, flight controls, structures) must be modeled in a common framework to catch interface issues early. As Boom implemented, a single 3D model shared across suppliers ensures “fits-right-first-time” and traceability of requirements [16†L80-L88] . Lockheed should incorporate model-based verification (virtual integration) to reduce physical prototyping.
- **Supplier Network:** Overture requires specialty suppliers (e.g. GE Aviation for engines, Collins or Honeywell for avionics, Spirit/Aero for wings, etc). A lean, geographically distributed supply chain (like F-35’s) will be needed for aerostructures, avionics, hydraulics and landing gear. LM’s

incumbent supply base (Alcoa/Constellium aluminum parts, Toray composites, Spirit F-22 fuselage) can be tapped, but new capabilities (automated composite layups at scale) might need partner development. Boom partners (Aernnova, Safran, Latecoere, Eaton) highlight that top-tier aerospace contractors can be integrated – Lockheed’s role would be systems integrator and final assembly.

- **Certification Data and Configuration Control:** Supersonic certification will require vast test data. Lockheed must set up comprehensive data management (product lifecycle management – PLM) systems from the outset, linking design requirements to analysis results and test data (as Boom did with ENOVIA/CATIA) 【16†L80-L88】. Strong configuration control is vital to meet DO-178/DO-254 (software/hardware) and FAR 25 subsystems tests.

7. Sustainment and Support

- **Maintenance Philosophy:** To avoid Concorde’s fate of high maintenance costs, Overture should aim for a high dispatch reliability. Engine maintenance intervals must balance thrust output with robustness; building on commercial turbofan technology (with longer overhaul periods than fighter engines) will help. Lockheed should plan for on-wing engine life similar to current narrowbodies (20,000+ hours). Composite structures generally reduce corrosion and fatigue issues, but require specialized repair techniques (Lockheed’s experience with the F-22 can be applied).
- **Supply and Logistics:** A worldwide spares network must be planned. Given limited fleet size, parts must be highly reliable and modular. Standardization (e.g. Line Replaceable Units for avionics) and use of common components (hydraulic pumps, electronics) from existing airliners can lower costs. Lockheed’s sustainment programs (like the F-35’s Global Support solution) provide a model: predictive maintenance analytics and managed spares pools can ensure fleet readiness.
- **Training and Operations:** Crew training (for high-altitude, high-Mach operations) will be extensive. Lockheed should work with airline partners (United, etc) to develop simulators early. Unique support (e.g. fuel monitoring for SAF, boom operation planning) will be part of the operations manual.

8. Regulatory & Certification Strategy

- **International Coordination:** Since supercruising and booms impact national policies, Lockheed must maintain liaison with ICAO (supporting supersonic CATE proposals) and FAA/EASA. Data from X-59 flights and XB-1 “boomless” flights provide a basis for new flight-test procedures (as ICAO is defining 【8†L333-L342】). Lockheed should advocate for a sonic-boom certification metric (as ICAO CAEP is developing) and ensure compliance with any new Annex 16 standards under development.
- **Airworthiness Certification:** The type certification plan must address novel issues: e.g. supersonic stall behavior, high-altitude flight physiology, and noise/thermal environments. Special conditions (like for SST flying at Mach>1) will be required. Lockheed can draw on FAA’s precedent (Concorde and SST experimental licenses) as guidance, but must align with new noise standards.

- **Environmental Certification:** Given the CO₂ and NO_x concerns, demonstration that 100% SAF use does not introduce new hazards (e.g. fuel system compatibility, material compatibility) will be necessary. Lockheed should coordinate with ICAO's Carbon Offsetting and Reduction Scheme (CORSIA) and U.S. emissions trading schemes to ensure Overture's operations earn credits.

9. Phased Roadmap, Risks and Cost Estimates

- **Phase 1 (2023–2025) – Concept and Preliminary Design:** Define mission requirements (seat count, range, speed). Conduct conceptual trade studies (e.g. Mach 1.7 vs 2.0). Engage NASA/FAA for preliminary data needs. Develop wind-tunnel and CFD models of candidate designs. *Milestones:* requirement set, trade study reports, baseline configuration frozen. *Risks:* Regulatory uncertainty; key supplier availability; higher-than-expected development cost.
- **Phase 2 (2025–2027) – Detailed Design and Prototype Development:** Finalize aerodynamic design, select or begin engine development (Boom's Symphony partnership with Kratos/GE suggests an in-house engine program). Build high-fidelity demonstrator (e.g. XB-1 or partial-scale plane). Begin iterative engine-inlet testing, materials fatigue tests. *Milestones:* Demonstrator first flight, preliminary propulsion prototype test. *Risks:* Engine R&D overruns; structural test failures; funding delays.
- **Phase 3 (2027–2029) – Flight Testing and Certification:** Complete full-scale prototype build and flight-test expansion to Mach cruise. Measure acoustic signature and refine flight controls. Compile certification data (performance, noise, safety). *Milestones:* Prototype flights at M1.7, noise certification campaigns, type-certification submission. *Risks:* Sonic boom still above threshold; unanticipated performance shortfall; certification standard changes.
- **Phase 4 (2029–2032) – Production and Entry Into Service:** Establish final assembly line ("Superfactory"), ramp up composite fabrication, secure airline orders. Implement maintenance support network. *Milestones:* Production readiness, first production flight, entry into commercial service. *Risks:* Market demand lower than forecast; production quality issues; supply chain bottlenecks.
- **Cost Estimate:** Past studies (Boom's estimates) place SST R&D and certification in the multi-billion-dollar range. One analysis put Overture's development+certification at ≈\$6 billion [38†L1-L4]. Lockheed should budget on this scale or higher (e.g. \$8–10B to include extra contingencies). Phasing investment through government contracts (e.g. NASA's X-59/NASA projects) and U.S. Air Force (like their Overture-AF1 deal) can mitigate risk.
- **Risk Mitigation:** *Technical:* Invest in breakthrough noise-reduction (e.g. laminar flow control, Boomless cruise modes). *Regulatory:* Partner with NASA/FAA to ensure design meets emerging rules. *Financial:* Secure anchor customers (e.g. major airlines or military variants) early. *Schedule:* Build on existing technology (use proven composite processes, adapt existing turbofan core) to shorten lead times.

10. Conclusion

Research indicates that a viable supersonic transport is technologically within reach if strict attention is paid to integration of advanced propulsion, aerodynamics and materials. Key enabling factors include: a low-noise engine (no afterburner) with high efficiency [35†L43-L51]; an airframe optimized by CFD for minimal wave drag [17†L68-L77]; and use of composites/titanium to save weight and tolerate heating

【25†L764-L772】 【31†L42-L50】 . Compliance with new international noise/emission standards will be critical; Lockheed must leverage data from the X-59 and XB-1 programs to validate quiet-supersonic designs 【6†L355-L363】 【35†L55-L63】 . By following a rigorous systems-engineering approach – integrating digital design, phased testing, and robust risk management – Lockheed Martin can chart a credible path to certifying a next-generation supersonic airliner in the late 2020s, balancing performance, cost and sustainability.

Sources: Government and industry reports (NASA, FAA, ICAO), Boom Supersonic and Rolls-Royce publications, and technical literature have been reviewed and are cited throughout 【4†L84-L93】 【15†L323-L332】 【35†L43-L51】 【25†L764-L772】 【27†L66-L70】 【37†L161-L164】 【38†L1-L4】 .
